

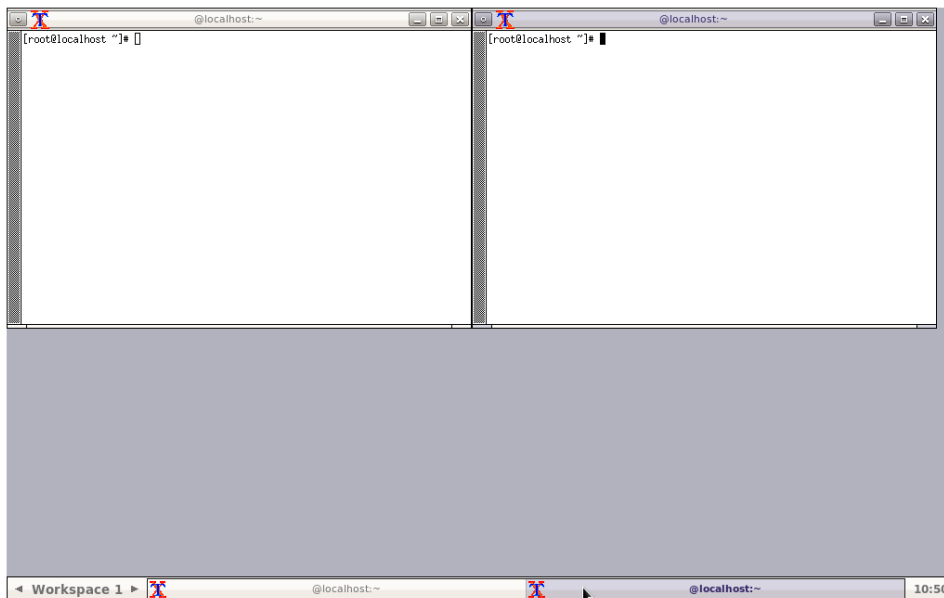
Session 7: File Systems

On completion of these activities, you should be able to:

- Use the **ls** command to list and understand information about file names and the i-nodes to which they refer.
- Use the **du** command to find out how much disk space is being used and where.
- Read a text file in C programming language.

Task 0

Start a UNIX environment in JSLinux (<https://bellard.org/jslinux/>). Select the Fedora X-Windows environment that should look something like the figure below.



Task 1

Create a new directory `session7` in your home directory. Use the **cd** command to move into `session7`. Use the **pwd** command to verify that your current directory is now `/root/session7`. Use the command **ls -ail** to show the current contents of `session7`. You should get a response similar to this:

```
[root@localhost session7]# ls -ail
total 8
229165 drwxr-xr-x 2 root root 37 Mar 18 10:52 .
      75 dr-xr-x--- 4 root root 352 Dec 26 2020 ..
[root@localhost session7]# █
```

The first line shows the total amount of space (in KiB) occupied by the files in the directory. The other two lines show information about the two directory entries `.` and `..`:

- The number on the left is the i-node number;
- the number before the user name is the i-node reference count, which is the number of references to the i-node which currently exist in the entire filesystem;
- the two names (both `root` in this example) are the names of the user and group, respectively, which own the file;
- the number before the date is the size (in bytes) of the file;
- the date/time shows when the file was most recently modified.

Now create a new file called *thingOne* in the directory (you could use an editor or the `touch` command to do this, for example). Run **ls -ail** again.

Now run this command: **ln thingOne thingTwo**. In some cases, you may need to be a superuser so you will need **su** or **sudo**.

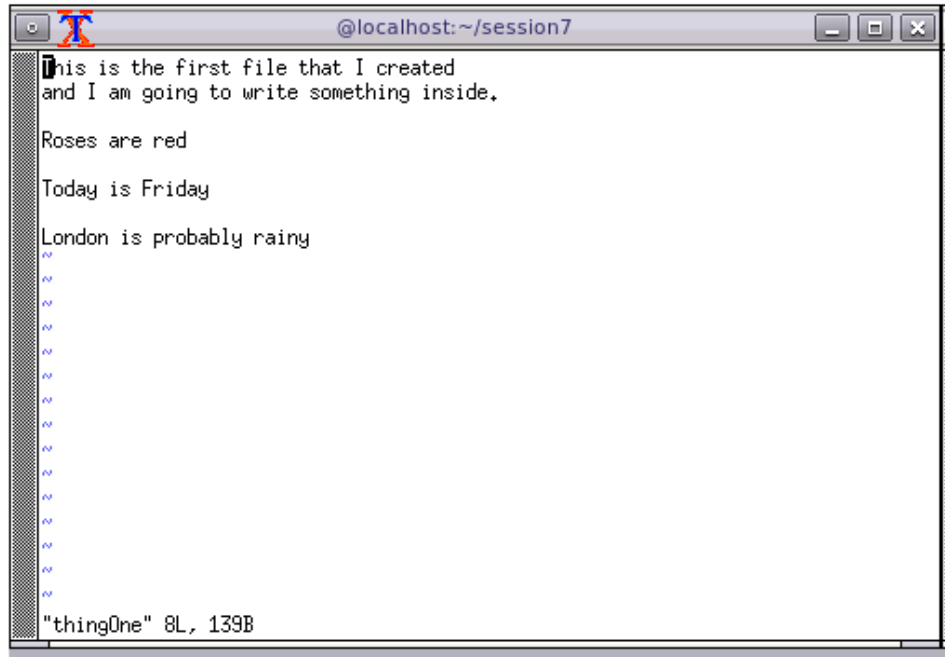
Run **ls -ail** again. What do you notice about the i-node numbers and reference counts? How do you explain this?

Task 2

Edit one of the files, you can do this with the command **vi**:

```
vi thingOne
```

To start writing you need to enter the edit mode by pressing **i** (for insert). Once you finish, you can save and exit by pressing **escape** and then **shift – zz**.



Now open the second file

```
vi thingTwo
```

Does this match what you were expecting?

Task 3

At some point, you will need to find out where all your disk space has gone. A basic tool for this is the **du** command. Make sure that your current directory is the same as the previous task /session7 (check with **pwd**; move with **cd** if necessary). Now enter:

```
du -h -c .
```

This asks for the total amount of space used within each file and directory in the current directory (.) presented in human-friendly units (-h) together with a grand total (-c) at the end. If you just want totals without all the details, add the summary option -s.

Do the same thing again but this time starting one level up, then repeat once more in the root (/) of the whole file system (**Warning: this may take a while...**).

Task 4 [optional]

Try running the sequential reading of a file as shown during the lecture. A file called readF.c is available on Moodle together with a snippet of Alice in Wonderland in alice.txt.

Remember that to upload the file you need to click the arrow at the bottom left:



Compile the file and run it

```
gcc -o readFile readF.c
./readFile
```

Change the size of the buffer and try again.